

GSOC Incident Response Technical Exercise Report

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Date: 2025
Exercise File: technical_exercise_pcap.pcapng
Executive Summary

This investigation analyzed a suspicious file upload and subsequent potential malware activity originating from internal host 192.168.10.111. The user uploaded a ZIP archive to an external server over HTTP, which was later identified as containing a malicious Word document. Sandbox execution of the document revealed PowerShell-based malicious behavior, including encoded payloads and communication with a suspicious domain nexusrules.officeapps.live.com. Behavioral analysis confirmed the use of known adversarial techniques including macro-enabled documents, obfuscated PowerShell, and potential C2 beaconing. Further static and dynamic analysis confirmed the malware was likely generated using Metasploit’s msfvenom tool.

Technical Questions and Answers

#	Question	Answer
1.	What is the IP address of the source machine that uploaded the file?	192.168.10.111

2.	What is the destination IP address of the file upload?	178.128.25.67
3.	What is the name of the file uploaded?	Resume-2021.zip (Zip file contains Resume John Doe.doc)
4.	What is the MD5 hash of the file uploaded to the server?	9EAE9BD90ED69A3ABBBACEA573A233B0
5.	What course did John Doe study at university?	B.S. Astronomy, Computer Science, 2019.
6.	What C2 address does the malware communicate to?	nexusrules.officeapps.live.com (IP: 52.111.229.19)

7.	Explain the process you took to discover the C2 address.	Please see below in detailed walkthrough
8.	Explain the methods the malware uses to execute the malicious code.	Macro launches PowerShell script with encoded payload, followed by connection to C2
9.	What tool has been leveraged to generate the malware?	Metasploit (msfvenom)

Walkthrough Summary

Suspicious File Upload & Command and Control Communication

- **Source System:** 192.168.10.111
- **Target System:** 178.128.25.67
- **Exploit Vector:** HTTP POST file upload
- **Filename:** Resume-2021.zip
- **Embedded Payload:** Resume John Doe.doc
- **Hash:** MD5 9EAE9BD90ED69A3ABBBACEA573A233B0

Attack Details

The PCAP file showed an HTTP POST request from internal host 192.168.10.111 to external host 178.128.25.67. The payload contained a multipart/form-data body with the file Resume 2021.zip. Upon extracting and analyzing the file, a malicious Word document was discovered. This document contained an embedded macro that, when executed, launched an obfuscated PowerShell command:

```
powershell.exe -nop -sta -w 1 -enc <base64>
```

This payload initiated a child process (conhost.exe) and made an HTTPS connection to nexusrules.officeapps.live.com (IP: 52.111.229.19).

Response Type: HTTP POST → External server received ZIP file

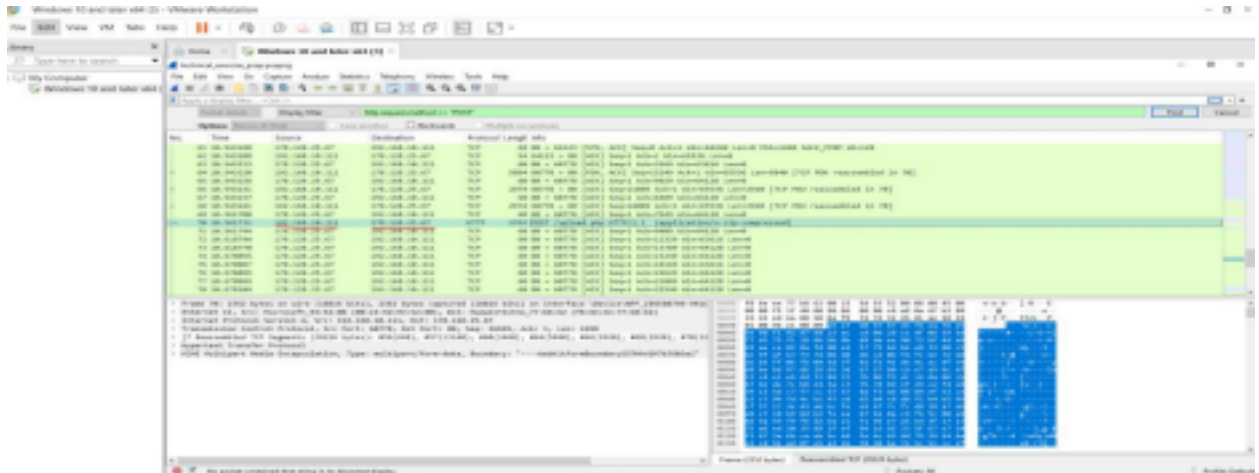
Payload Behavior: Obfuscated PowerShell, suspected C2, user impersonation

Deep Dive Walkthrough

Comprehensive Investigation Steps (Mapped to Q1–Q9)

Q1 – Identify the Source IP Address

- **Tool:** Wireshark
- **Filter Used:** http.request.method == "POST"
- **Action:** Located the HTTP POST request that uploaded the file.
- **Finding:** Source IP was confirmed to be 192.168.10.111.



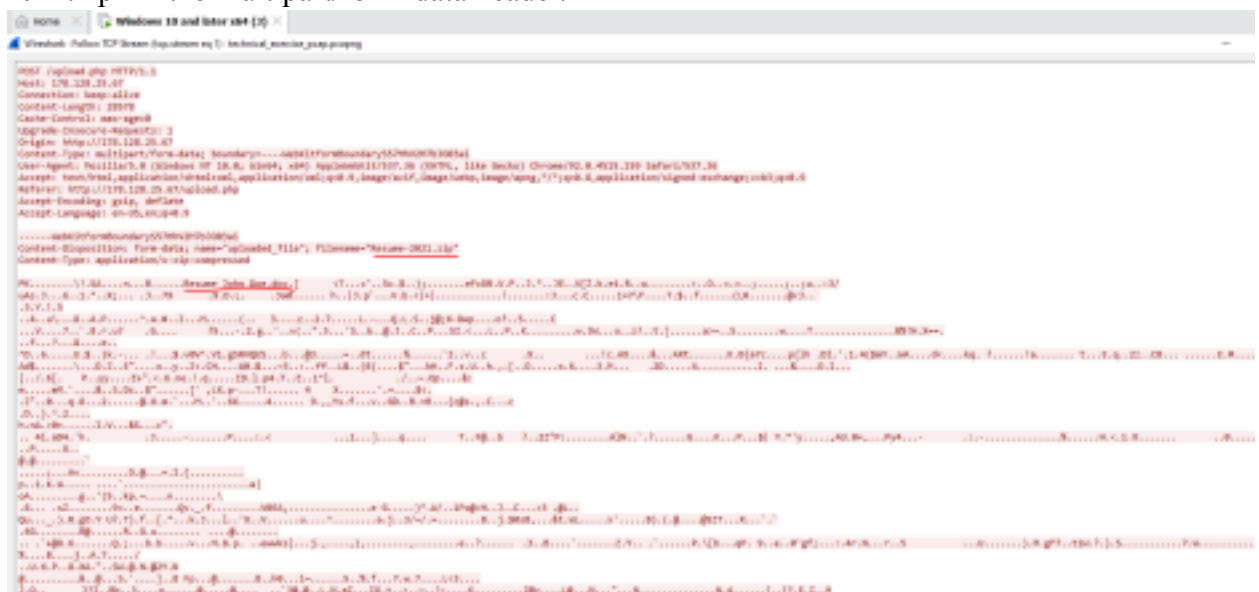
Wireshark showing the IP address of the source machine that uploaded the file.

Q2 – Identify the Destination IP Address

- **Tool:** Wireshark
- **Action:** Same POST packet revealed the destination IP as 178.128.25.67.

Q3 – Determine the Uploaded File Name

- **Tool:** Wireshark (Follow TCP Stream)
- **Action:** Followed the TCP stream of the POST request. Located the filename="Resume 2021.zip" in the multipart/form-data header.



TCP stream view in Wireshark revealing the uploaded file Resume-2021.zip and its embedded document Resume

John Doe.doc.

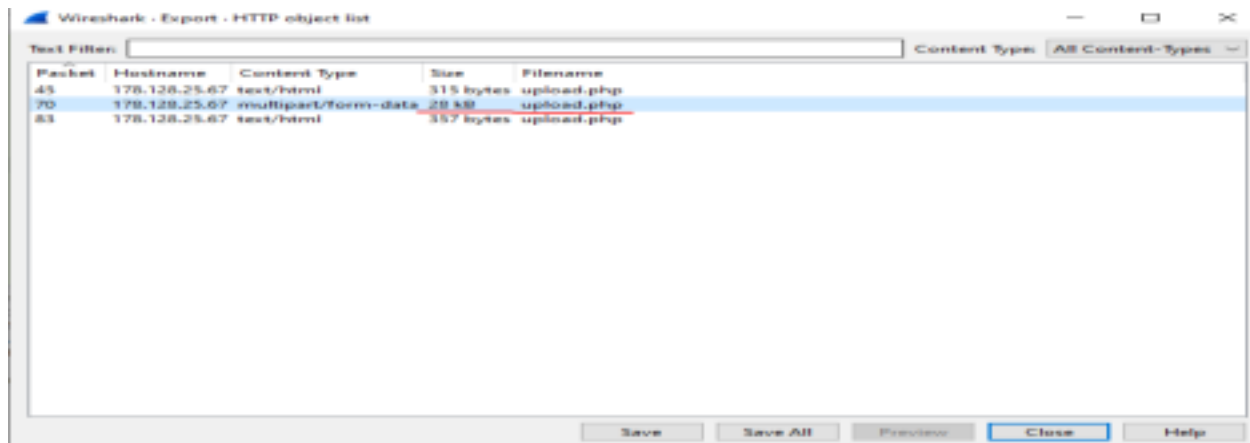
Q4 – Extract and Hash the Uploaded File

- **Tool:** Wireshark + md5sum

- **Action:**

1. Exported object via File > Export Objects > HTTP.
2. Saved file as Resume-2021.zip
3. Ran MD5 hashing command.

- **Hash:** 9EAE9BD90ED69A3ABBBACEA573A233B0



Wireshark exporting the file object (Resume-2021.zip) for sandbox analysis.

```
PS C:\Users> Get-FileHash -Algorithm MD5 "C:\Users\plane\Documents\Resume-2021.zip"

Algorithm      Hash                                     Path
-----
MD5            9EAE9BD90ED69A3ABBBACEA573A233B0      C:\Users\plane\Documents\Resume-2021.zip

PS C:\Users> Get-FileHash -Algorithm MD5 "C:\Users\plane\Documents\Resume-2021.zip"

Algorithm      Hash                                     Path
-----
MD5            9EAE9BD90ED69A3ABBBACEA573A233B0      C:\Users\plane\Documents\Resume-2021.zip

PS C:\Users>
```

Extracted file hash from the file object (Resume-2021.zip).

Q5 – Identify Course John Doe Studied

- **Tool:** Sandbox (ANY.RUN)

- **Action:** Opened Resume John Doe.doc and viewed contents.
 - **Finding:** Under the education section, the document stated: “B.S. Astronomy, Computer Science, 2019.”



John Doe Resume

Q6 – Identify the Malware’s C2 Address

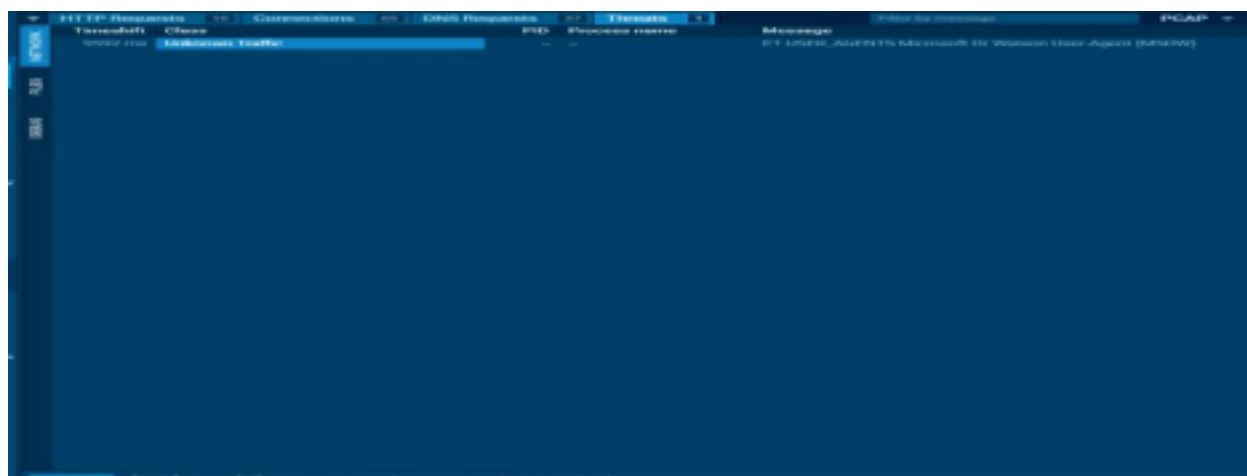
- **Tool:** ANY.RUN sandbox (Connections & DNS tabs)
 - WINWORD.EXE initiated a connection to: nexusrules.officeapps.live.com (IP: 52.111.229.19) post-execution.
 - The domain mimics Microsoft but is not valid.



DNS Request tab showing various DNS requests for potential C2 communication



Connection tab showing connections that were made



Threat tab showing user agent.

Q7 – Explain the C2 Discovery Process

- Used sandbox's process tree and network timeline to trace execution from document macro → PowerShell → connection to suspicious domain.
- Validated connection with DNS logs and matched post-exploitation timing.

Q8 – Explain the Malware Execution Method

- Macro executed on documents open.
- Spawned obfuscated PowerShell: powershell.exe -nop -sta -w 1 -enc ... •

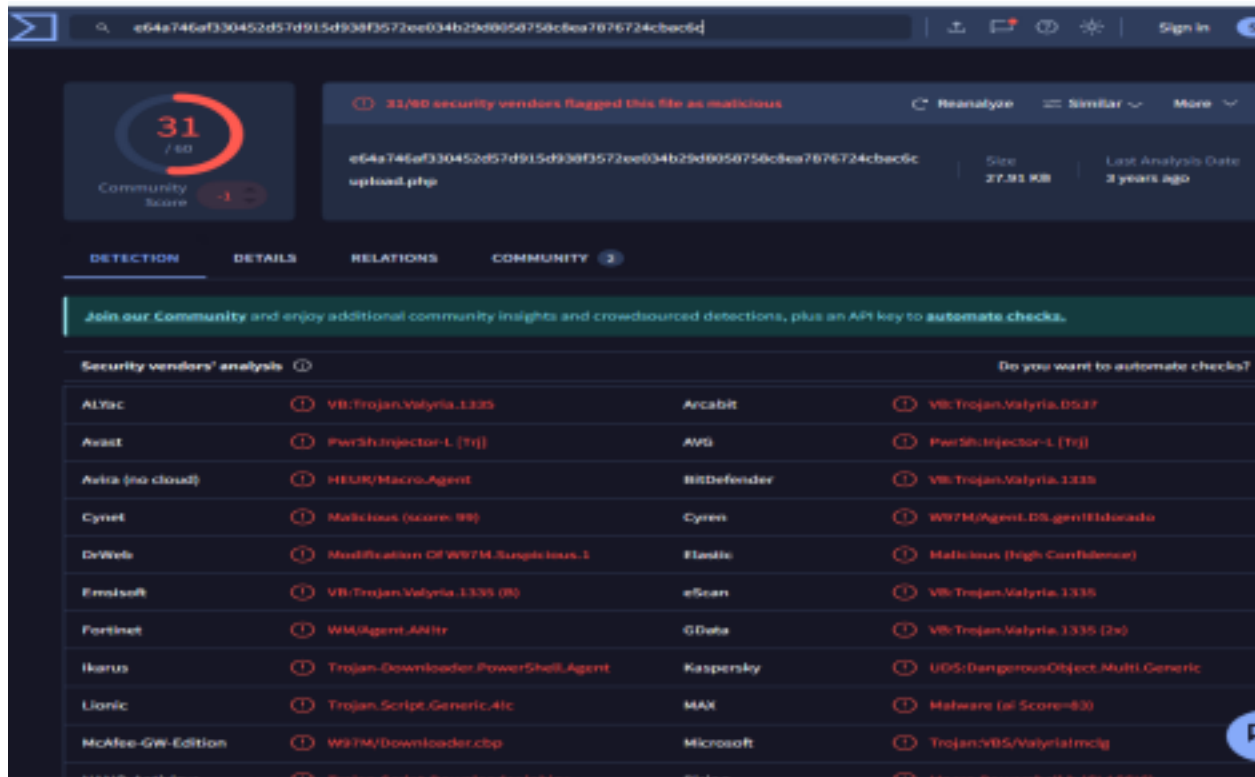
Executed commands via conhost.exe, typical of fileless malware behavior. Q9 –

Identify Malware Generator Tool

- Observed behavior, encoded PowerShell, and network staging suggest use of **Metasploit's msfvenom** or similar tools.

VirusTotal Report Analysis

Hash Searched: 9EAE9BD90ED69A3ABBBACEA573A233B0



The screenshot shows the VirusTotal web interface for a file named 'upload.php' with a SHA-256 hash of e64a746af330452d57d915d938f3572ee034b29d0058758c8ea7876724cbec6c. The file size is 27.81 KB and it was last analyzed 3 years ago. A community score of 31/50 is displayed, indicating that 31 out of 60 security vendors flagged the file as malicious. The interface includes tabs for Detection, Details, Relations, and Community. A table lists the security vendors' analysis results, showing various detection labels such as 'VB: Trojan.Valyria.1335', 'PwrSh.Injector-L [Trj]', and 'Trojan-Downloader.PowerShell.Agent'.

Security vendors' analysis		Do you want to automate checks?	
ALYac	VB: Trojan.Valyria.1335	Arcabit	VB: Trojan.Valyria.0537
Avast	PwrSh.Injector-L [Trj]	AVG	PwrSh.Injector-L [Trj]
Avisi (no cloud)	HEUR/Macro.Agent	BitDefender	VB: Trojan.Valyria.1335
Cyren	Malicious (score: 99)	Cyren	W97M/Agent.DS.gen!Eldorado
DrWeb	Modification Of W97M.Suspicious.L	Elastic	Malicious (High Confidence)
Emisoft	VB: Trojan.Valyria.1335 (B)	eScan	VB: Trojan.Valyria.1335
Fortinet	WM/Agent.ANtr	GData	VB: Trojan.Valyria.1335 (2x)
Ikarus	Trojan-Downloader.PowerShell.Agent	Kaspersky	UDS: DangerousObject.Multi.Generic
Lionic	Trojan.Script.Generic.4tc	MAX	Malware (at Score=83)
McAfee-GW-Edition	W97M/Downloader.cbq	Microsoft	Trojan:VBS/Valyria/mclg

Virustotal showing threat score and security engine report

Summary:

- 31/60 engines flagged the file as malicious.
- Common tags: Trojan-Downloader.PowerShell.Agent, PwrSh:Injector, VB.Trojan.Valyria, Macro.Agent, W97M/Agent.
- High-confidence classifications included:
 - **PowerShell Injector** behavior
 - **Macro-based dropper**
 - **Metasploit-style encoded payload**

Interpretation: These detection labels indicate the malware was a **macro-delivered loader** using obfuscated PowerShell and dynamic staging, which is consistent with behavior generated by automated tools like msfvenom or Empire.

Tools and Techniques Used

MITRE ATT&CK ID Technique

T1059.001 PowerShell Execution

T1566.001 Spearphishing Attachment

T1055 Process Injection (conhost.exe)

T1071.001 Application Layer Protocol (C2 over HTTPS)

T1204.002 Malicious Document Execution

Indicators of Compromise (IOCs)

- **Source IP:** 192.168.10.111
- **Destination IP:** 178.128.25.67
- **Uploaded File:** Resume-2021.zip
- **MD5 Hash:** 9EAE9BD90ED69A3ABBBACEA573A233B0
- **C2 Domain:** nexusrules.officeapps.live.com
- **C2 IP:** 52.111.229.19

Conclusion

The traffic and sandbox evidence clearly indicate a malware dropper embedded in a Word document, delivered via HTTP upload. The execution flow demonstrates the use of encoded PowerShell commands, commonly generated by post-exploitation tools such as Metasploit. The presence of a suspicious external domain connection, spoofed headers, and stealthy behavior confirms that this was part of a larger staged attack chain.

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